

Quiz 2

Section BSE-4B

Question

A university computer lab has only **3 printers** connected to a central printing system. Multiple students send print requests simultaneously from different computers.

Each student is represented by a separate thread. Since printers are shared resources, the system must ensure proper synchronization so that:

1. Only one student can use a printer at a time.
2. No two students print on the same printer simultaneously.
3. If all printers are busy, incoming students must wait.
4. After printing, the printer becomes available for another student.
5. Use **mutex locks** to synchronize access to printers.

Tasks

1. Write a multithreaded C program using **POSIX threads (pthreads)**.
2. Simulate multiple students sending print jobs.
3. Display:
 - Which student gets which printer.
 - When printing starts and finishes.
4. Use synchronization to avoid race conditions.
5. Explain what problem occurs if synchronization is not used.